Quadratic Refinements in Normal Play: Realization and the Observation Boundary

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Abstract

Classical coin-turning games decompose into nim heaps, and their P-positions are therefore $\mathbb{F}_2$-linear; the same linear-kernel phenomenon underlies anncodes, Mock Turtles, and Mogul. We construct, to our knowledge, the first uniform normal-play family that strategically forces a genuinely quadratic P-set and proves the matching observation boundary. The theorem is not Gold-specific: for every finite-dimensional $\mathbb{F}_2$-space, every alternating form B, every quadratic refinement Q of B, and every loaded input x, the root is P exactly when $Q(x) = 0$.

The statement includes its access class. The arena obeys F1–F3/N1–N2: its statics are refinement-blind, each transition makes at most one weight-one refinement observation, the semantics is ordinary normal play, and exactness is uniform over the full torsor of refinements. A public Witt frame reduces the polar support to a matching; singleton-weighted source pairs transport the diagonal data; a FIFO strategy forces every potential pair to overlap; and a phase-aware terminal claim compiles the resulting charge. The phase ϕ is the mover field, so the compiled arena is partizan. Whether an impartial uniform realizer exists remains open.

The Gold–Arf family $Q_a(x) = \mathrm{Tr}(x\,x^{2^a})$ is the motivating game-built instance: nim-product, Frobenius, trace, and XOR source its coefficients, and the Arf invariant with radical data gives the exact second-player bias. For the general theorem, transcript stability forces the observed directions to span x; hence weight-w observations require at least $\mathrm{wt}(x)/w$ queries. This is attained by the construction at $w = 1$; no matching upper bound is claimed for $w > 1$. The Lean supplement checks five independent ingredients of the realization and lower bound, rather than one end-to-end arena theorem. A separate kernel-checked padding result shows why extensional winning-fork tests cannot certify naturality.

1 Introduction and scope

The linear ceiling of established coin-turning constructions is unusually rigid. Ferguson's decomposition expresses every such terminating ruleset as a disjunctive sum of nim heaps [17]; Fraenkel's annihilation games consequently produce linear error-correcting codes ("anncodes") [18]. Mock Turtles and Mogul fit the same pattern, the latter recovering the binary Golay code. Thus their P-sets are kernels of $\mathbb{F}_2$-linear maps. The present result crosses that ceiling: it forces $\{Q = 0\}$ for every quadratic refinement of every alternating form, while proving exactly how much refinement data any transcript-stable rule must expose.

The access condition is essential. Any prescribed finite subset can be made the P-set of an ad hoc acyclic digraph, so a realization theorem without a uniformity and information contract is vacuous. Definition 2 fixes that contract as F1–F3/N1–N2. The headline construction meets it with $(w_0, c) = (1, 1)$: public B-data may organize the board, but a transition sees at most one singleton value of the refinement. It works uniformly for all Q, not merely for the Gold trace forms that led to it.

The closest neighboring constructions have different input semantics. Johnson's input-setting games let the players choose the Boolean input, whereas here x is loaded data and play must reveal its quadratic value under a fixed local rule [23]. Brandenburg's algebraic games play directly on presented groups and rings rather than compile a quadratic predicate [4]. The invariant-games program asks whether a position-independent subtraction rule has a prescribed P-set [14]; our Definition 2 is an explicit stateful analogue of that invariance question, allowing FIFO memory while fixing the loading, statics, and oracle budget. Literal move-digraph invariance is too rigid here (Theorem E), so the positive theorem uses covariant public preprocessing instead.

There is also a classical graph-theoretic ancestor of the charge mechanism. The Cohn–Lempel equality, Bouchet's isotropic systems, and the modern interlacement-matroid formulation show how chord intersections and interlacement parity produce quadratic forms over $\mathbb{F}_2$ [8, 3, 5]. What is new here is not that overlap parity is quadratic, but that normal play can force the required overlaps strategically under a bounded-access contract.

The arena is partizan only through its mover bit ϕ, which assigns the terminal claim to the correct seat. An impartial uniform realization with the same access discipline is open. The Lean supplement checks five independent ingredients: the diagonal trace identities, the matching FIFO strategy, the adapted-basis expansion, the phase-aware claim compiler, and the transcript-span bound. It also checks the separate fork-padding no-go. As Section 11 records, these are ingredient theorems; no single Lean statement constructs the complete arena.

A short partizan game is built recursively as $\{L \mid R\}$, and games form a partially ordered abelian group under disjunctive sum [10, 9]. They do not form a ring: Conway multiplication is defined on the number subclass, not on arbitrary games. Since a Clifford algebra needs a commutative scalar ring, the phrase "Clifford over games" has to be interpreted through scalar subworlds:

world	algebraic structure	role here
No	real-closed field, char 0	ordinary real Clifford theory with surreal scalars
No$[i]$	algebraically closed field, char 0	ordinary complex Clifford theory
finite nimber subfields	finite fields of char 2	characteristic-2 Clifford and Arf experiments

The implemented `Nimber` scalar is finite: `u128` realizes $\mathbb{F}_{2^{128}}$, containing the nimber subfields $\mathbb{F}_{2^m}$ for $m = 1, 2, 4, \ldots, 128$. It is not the whole proper-class field On_2.

2 The char-2 Clifford datum

In characteristic 2, a quadratic form is not determined by its polar form. For basis vectors the Clifford relations are

$$e_i^2 = q_i, \qquad e_ie_j + e_je_i = b_{ij}.$$

The polar form is alternating, so $b_{ii} = 0$, but q_i can be nonzero. A faithful characteristic-2 implementation must store q and b separately. This is the main implementation discipline behind the nimber backend, and – as Section 7 makes precise – it is also the group-theoretic content of the whole note: q and b are the independent central data (squares and commutators) of an extraspecial-type extension.

For a nonsingular quadratic form over $\mathbb{F}_2$ with symplectic basis (a_k, b_k), the Arf invariant is

$$\mathrm{Arf}(Q) = \sum_k Q(a_k)Q(b_k) \in \mathbb{F}_2.$$

It classifies nonsingular binary quadratic spaces at fixed dimension. Over a finite characteristic-2 field, the corresponding value lies in $F/\wp(F)$, with $\wp(y) = y^2 + y$; for finite fields this quotient is read by the field trace to $\mathbb{F}_2$.

Remark 1. Ovsienko [29] identifies a precise bridge between Dickson's classification of quadratic forms over $\mathbb{F}_2$ and the classical classification of real Clifford algebras. We use that paper as motivation for the Arf/Clifford connection, not as a blanket classification theorem for all Clifford algebras over finite nimber fields. In the code, rank and radical data are reported alongside Arf, and degenerate forms are not collapsed to a single Arf bit.

The small finite-field check used throughout is:

$$q = (*2, *2),\ b_{01} = *1 \quad \text{has Arf } 1,$$

whereas

$$q = (*2, *3),\ b_{01} = *1 \quad \text{has Arf } 0.$$

The polar entry is part of the form.

3 Gold forms from game operations

Nim-addition is XOR, and XOR is the disjunctive sum of impartial game values. Nim-multiplication is also game-theoretic: Conway's Turning-Corners product [10, 26] has the mex recurrence

$$x \otimes y = \operatorname{mex}\left\{(i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \le i < x,\ 0 \le j < y\right\}. \tag{1}$$

The repo implements this recurrence directly for small arguments and checks it against the fast algebraic nim product.

Thus the following operations are game-realizable at the level of nimber values:

$$x \oplus y, \qquad x \otimes y, \qquad x \mapsto x^2 = x \otimes x, \qquad \mathrm{Tr}(x) = x + x^2 + \cdots + x^{2^{m-1}}.$$

For $F = \mathbb{F}_{2^m}$ and $a \ge 1$, define the Gold trace form

$$Q_a(x) = \mathrm{Tr}\big(x\, x^{2^a}\big) = \mathrm{Tr}\big(x^{1+2^a}\big)\,.$$

Since Frobenius $x \mapsto x^{2^a}$ is additive, Q_a is a quadratic form over $\mathbb{F}_2$. In the nimber backend, it is also a composite of the game operations above. Throughout, "the instance (m, a)" means Q_a on $\mathbb{F}_{2^m}$, and "(m, a, λ)" the scaled component $Q_{\lambda,a}(x) = \mathrm{Tr}(\lambda x^{1+2^a})$.

Observation 1. *The Gold forms are not merely defined on a field of game values. In the finite nimber fields tested here, they are built from nim/game operations: Turning Corners for multiplication, diagonal Turning Corners for Frobenius squaring, and disjunctive sum for XOR and trace.*

This observation is modest but useful: it gives a concrete quadratic form at the interface between impartial-game arithmetic and characteristic-2 quadratic-form theory.

3.1 Rank check

The polar rank of Gold forms is known [22]. In the tested regime $m = 2^k$ and $1 \le a \le k$, the general formula

$$\operatorname{rank} B_{Q_a} = m - \gcd(2a, m)$$

specializes to $m - 2\gcd(a, m)$ because $m/\gcd(a, m)$ is even. The classifier reproduces the general formula in all tested cases. Representative rows are shown in Table 1.

field	a	Q	Arf	type	rank	formula
$\mathbb{F}_{2^4}$	1	$\mathrm{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\mathrm{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\mathrm{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\mathrm{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\mathrm{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\mathrm{Tr}(x^3)$	1	O^-	30	30

Table 1: Representative output of `experiments/trace_form_arf.py`. The script checks all 15 cases with $m = 2, 4, 8, 16, 32$ and $1 \le a \le \log_2 m$.

Most of these forms have a nontrivial radical. The statement is therefore not "nondegenerate Gold form" but "positive-rank quadratic form with a nonsingular core whose Arf value is computed after symplectic reduction."

3.2 A broader trace family

The coefficient-1 Gold form is only one trace monomial. A standard trace description of the quadratic part of Boolean functions over $\mathbb{F}_{2^m}$ uses terms

$$\mathrm{Tr}(c_i x^{1+2^i})$$

plus affine terms and, in even degree, a middle half-trace term. The monomials displayed here are still game-operation-built in the nimber backend: $c_i x^{1+2^i} = c_i \otimes x \otimes x^{2^i}$, followed by trace/XOR.

The current implementation does not claim to package every Boolean quadratic function. The probe `experiments/gold_family_survey.py` deliberately omits the middle term and affine bookkeeping. What it does check, exhaustively over $\mathbb{F}_{256}$, is that scaled Gold components

$$Q_{\lambda,a}(x) = \mathrm{Tr}(\lambda x^{1+2^a})$$

already reach nondegenerate (bent) forms for many λ. For APN Gold exponents in that field, the observed count matches the classical $2(2^m - 1)/3$ bent-component count. This matters for the game question because a bent form has trivial radical: the loopy radical route collapses to $\{0\}$, and the frame-blind symplectic no-go applies without a degenerate layer to hide in.

Remark 2 (a vacuous small target)**.** At $m = 4$, $a = 1$ the zero set $\{Q_1 = 0\} \subset \mathbb{F}_{16}$ is the subfield $\mathbb{F}_4$, an $\mathbb{F}_2$-*subspace* of dimension 2: the form is genuinely quadratic (polar rank 2), but its zero set is affine, so as a target for "is this P-set a genuine quadric" it is vacuous – a linear rule realizes it. The smallest honest unscaled instance is $(8, 1)$. The repository's quadric-fitting layer (`src/forms/quadric_fit.rs`) reports exactly this distinction (genuine polar rank versus affine flat), and every probe below that consumes a fitted quadric is read with that guard in mind.

4 Arf as win-bias

The relevant counting theorem is classical. If Q is nonsingular on $\mathbb{F}_2^{2r}$, then

$$\#\{x : Q(x) = 0\} = 2^{2r-1} + (-1)^{\mathrm{Arf}(Q)} 2^{r-1}.$$

For degenerate forms, the code uses the standard radical-adjusted count: an anisotropic radical balances the values, while an isotropic radical scales the bias of the nonsingular core.

The counting theorem becomes the game interpretation after Theorem 8:

> For weighted-source WITT-FIFO, P-positions are exactly $\{x : Q(x) = 0\}$. On a nonsingular core the Arf invariant is the sign of the bias between second-player wins and first-player wins; in the singular case the radical data supplies the standard adjustment.

Section 10 supplies the rule and proof that make this interpretation unconditional for the stated game.

field	a	Arf	rank	$\#\{Q = 0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Representative zero-counts from `experiments/arf_win_bias.py`. The bias is $\#\{Q = 0\} - 2^{m-1}$.

5 The linear ceiling

For a normal-play disjunctive sum of impartial games, the P-condition is linear in Grundy coordinates:

$$g_1 \oplus \cdots \oplus g_n = 0.$$

So normal-play sums give linear P-sets, not genuine quadrics. The coin-turning architecture – the very architecture whose product theorem defines the nim product – is subject to the same ceiling.

Theorem A (coin-turning linearity; standard math, machine cross-checked)**.** *Every Winning Ways coin-turning game has configuration P-set equal to the kernel of an $\mathbb{F}_2$-linear nimber-valued map. In particular no coin-turning P-set is a quadric of positive polar rank.*

Proof sketch. A coin-turning position is a finite set of heads; by the Winning Ways coin-turning theory [9], the Grundy value of a configuration is the XOR of the single-coin Grundy values $g(i)$ of its heads. The P-set is therefore $\{x : \bigoplus_{i \in x} g(i) = 0\}$, the kernel of the $\mathbb{F}_2$-linear map $x \mapsto \bigoplus_i x_i\, g(i)$. The repository cross-check (`experiments/gold/octal_attack.py`) verifies XOR-additivity by mex recursion, exhaustively over all 32,768 companion-family choices on 4 coins. □

Theorem A has a sharp classical complement: the linear ceiling is exactly where coding theory already harvests games.

Remark 3 (the lexicode shadow; standard math and interpretation)**.** Conway–Sloane lexicodes [11] are built by the greedy lexicographic rule – the mex rule – and the 1986 paper realizes them as the Grundy-value-0 positions of associated turning-game move structures; binary lexicodes are linear *because of* Sprague–Grundy theory (XOR-closure is a game theorem there, not a coding theorem). They include the Hamming codes and the length-24, $d = 8$ extended binary Golay code; over base 2^k they are closed under nim-addition, and they are linear at the Fermat-power bases 2^{2^k} – exactly the sizes at which nim-multiplication makes the ordinals below the base a field. Read together with Theorem A: lexicodes are the *floor* (rich linear codes naturally realized as Grundy-0 sets) and Theorem A is the *ceiling* (coin-turning realizes only linear P-sets); floor and ceiling meet at linear. A quadratic refinement cannot remain inside this disjunctive-sum ceiling: its obstruction is precisely the XOR-closure failure measured by the polar form. Weighted-source FIFO supplies the required quadratic realization by leaving the disjunctive-sum model.

For a quadratic form in characteristic 2,

$$Q(u + v) = Q(u) + Q(v) + B(u, v).$$

Thus for $u, v \in \{Q = 0\}$,

$$u + v \in \{Q = 0\} \iff B(u, v) = 0.$$

The polar form is exactly the obstruction to XOR-closure. The first two layers are game-built: XOR is disjunctive sum and B is built from nim products. The play layer supplied in Section 10 integrates that bilinear data into the quadratic zero set.

6 Routes and obstructions

6.1 Test benches

The repo contains several probes. They are useful for narrowing the target, but they are not a proof that no natural game exists.

- `fit_f2_quadratic` tests whether a subset of $\mathbb{F}_2^k$ is the zero set of a quadratic polynomial, and whether the result has nonzero polar rank (distinguishing genuine quadrics from affine flats; cf. Remark 2).
- A bounded misere quotient computer tests small quotient P-sets. The octal sweep checked 292 octal codes, heap cutoff at most 4, and found quotient orders $2, 6, 10, 12, 14$, with no elementary abelian 2-group of rank at least 2. Conditional statement C below shows this emptiness is forced, subject to its cited regularity hypothesis, rather than a sampling accident.
- `experiments/misere_kernel.py` checks the kernel obstruction on the smallest wild misere quotient R_8: the canonical group/kernel shadow is linear, while the genuinely misere P-element lies outside that vector-space part. Corollary 1 below shows the linearity is forced by commutativity.
- An interactive-kernel solver tests finite move graphs. It confirms that an ad hoc acyclic game can realize any subset, and that a rule referencing Q itself realizes $\{Q = 0\}$ tautologically. The tested B-coupled rules do not produce the Gold zero set.
- `examples/loopy_quadric.rs` tests cyclic move graphs with Draw sets. The symmetric B-coupled rule has Loss-set equal to the radical $R(B)$, so it reproduces the obstruction rather than the Gold zero set; this is the special case of Theorem G below, and its small $(4, 1)$ "hit" is the radical coincidence $|\{Q_1 = 0\}| = 4 = |R(B)|$ there.

- `examples/bent_route.rs` repeats route probes on a bent scaled Gold component. In that example, a B plus coordinate-frame rule reaches a quadric of the correct Arf class but not the specific Gold zero set; the naive local-field/spin-flip completion leaves the quadric variety.

Except for the frame-blind proposition below and the lettered theorems of Section 8, these are bounded or example-level negative results.

6.2 Frame-blind no-go

There is a clean obstruction for rules that see only a nondegenerate symplectic form and no frame.

Proposition 1. *Let B be a nondegenerate alternating form on $V = \mathbb{F}_2^{2r}$ with $r \geq 2$. If a finite normal-play game has position set V and a move relation invariant under $\mathrm{Sp}(B)$, then its P-set is not a nondegenerate quadric.*

Proof. Each element of $\mathrm{Sp}(B)$ is an automorphism of the move graph, so the retrograde outcome function is $\mathrm{Sp}(B)$-invariant. The P-set is therefore a union of symplectic orbits. For $r \geq 2$, the symplectic group is transitive on $V \setminus \{0\}$ [35], so the invariant subsets are only $\varnothing$, $\{0\}$, $V \setminus \{0\}$, and V. A nondegenerate quadric in dimension at least 4 has $2^{2r-1} \pm 2^{r-1}$ points, which is none of these. □

The hypothesis matters. Coordinate-dependent rules have already broken $\mathrm{Sp}(B)$ symmetry, so this proposition does not explain every negative probe. It explains only genuinely frame-blind B-only rules. Section 8 extends the obstruction far beyond this symmetric class.

6.3 The diagonal framing

Once a coordinate frame is allowed, a canonical-looking quadratic refinement of B is

$$Q_{\mathrm{frame}}(v) = \sum_{i<j} B(e_i, e_j) v_i v_j.$$

For the Gold polar forms tested in this repo, the Gold form decomposes as

$$Q_a(v) = Q_{\mathrm{frame}}(v) + \ell_{\mathrm{diag}}(v), \qquad \ell_{\mathrm{diag}}(v) = \sum_i Q_a(e_i) v_i.$$

In the tested genuinely quadratic Gold cases up to $\mathbb{F}_{2^{16}}$, Q_{frame} has Arf 0 and the diagonal term changes the form to the Gold Arf-1 class.

Remark 4. This is an experimental pattern for the Gold polar forms tested here, not a theorem about every alternating form plus every coordinate frame. Random nondegenerate alternating forms can give a zero-diagonal frame quadric of Arf 1.

The diagonal $q_i = Q_a(e_i)$ is therefore the named missing datum. Two results of the parallel run locate it more precisely.

Proposition 2 (game-native diagonal source for $a = 1$; implemented and tested)**.** *Let $m = 2^k$ and let $w \in \mathbb{F}_{2^m}$ be the XOR of the Fermat coins 2^{2^t} for $t = 1, \ldots, k-1$. Then the $a = 1$ Gold diagonal satisfies*

$$q_i^{(m,1)} = Q_1(e_i) = \mathrm{Tr}\big(\wp(w)\, e_i\big), \qquad \wp(w) = w \otimes w \oplus w,$$

i.e. the trace-dual element of the diagonal is $\wp(w)$. Verified at $m = 4, 8, 16, 32$ via the trace-pairing uniqueness of the dual element (`experiments/gold/gold_diag_probe.py`).

This gives a game-native, tower-uniform, commutative source for the $a = 1$ diagonal: one nim-squaring, one XOR, one trace – all inside the licensed operation chain of Section 3. The following theorem closes the same sourcing question for every exponent, including even a, and identifies the exact obstruction for a general scale.

For $c \in K_m = \mathbb{F}_{2^m}$, let $\lambda_{a,c}^{(m)}$ be the unique trace-dual of the scaled canonical-basis diagonal:

$$\mathrm{Tr}_m(\lambda_{a,c}^{(m)} e_i) = \mathrm{Tr}_m(c e_i \,\mathrm{Fr}^a(e_i)) \qquad (0 \le i < m). \tag{2}$$

Theorem 1 (sharp diagonal-source criterion)**.** *For every admitted power-of-two degree m and every $a \ge 0$,*

$$\mathrm{Tr}_m(\lambda_{a,c}^{(m)}) = \mathrm{Tr}_m(c), \qquad \lambda_{a,c}^{(m)} \in \wp(K_m) \Longleftrightarrow \mathrm{Tr}_m(c) = 0.$$

In particular, if $m = 2^k \ge 2$ and $c = 1$, there is a $w_a^{(m)} \in K_m$ such that

$$\lambda_{a,1}^{(m)} = (w_a^{(m)})^2 + w_a^{(m)}, \qquad Q_a(e_i) = \mathrm{Tr}_m(((w_a^{(m)})^2 + w_a^{(m)})e_i)$$

for every canonical basis coin e_i.

Proof. Put $e_0 = 1$ in the defining duality (2):

$$\mathrm{Tr}_m(\lambda_{a,c}^{(m)}) = Q_{a,c}(1) = \mathrm{Tr}_m(c).$$

The Artin–Schreier image of a finite characteristic-two field is exactly the absolute-trace kernel [27, Thm. 2.25]. This proves the equivalence. For the unscaled form, $\mathrm{Tr}_m(1) = m \bmod 2 = 0$ whenever the power-of-two degree satisfies $m \ge 2$. □

The one-line proof is basis-independent: it works in any basis containing 1. The stronger descent and recursion below are special to Conway's bit basis. Write $m = 2M$, $K_m = K_M(u) = K_M \oplus uK_M$, where $u = e_M$. Conway's Fermat-power rules give

$$\sigma(u) = u + 1, \qquad ue_j = e_{M+j}, \qquad \sigma = \mathrm{Fr}^M$$

[10, Ch. 6]. If $e_j^{*,(M)}$ denotes the trace-dual of e_j in K_M, then the trace-dual bit basis upstairs has the closed form

$$e_{M+j}^{*,(2M)} = e_j^{*,(M)}, \qquad e_j^{*,(2M)} = (1 + u)e_j^{*,(M)}. \tag{3}$$

Indeed, for $A, B, e \in K_M$ the relative trace $T = 1 + \sigma$ satisfies

$$T((A + uB)e) = Be, \qquad T((A + uB)ue) = (A + B)e. \tag{4}$$

Substitution of $A = 0, B = e_j^{*,(M)}$ and then $A = B = e_j^{*,(M)}$ gives the two dualities in (3), including the vanishing cross blocks.

Two consequences now take one line each. Let

$$d_0 = T(c), \qquad d_1 = T(cu \,\mathrm{Fr}^a(u)), \qquad L_s = \lambda_{a,d_s}^{(M)}.$$

The lower and upper diagonal blocks have duals L_0, L_1, so

$$\lambda_{a,c}^{(2M)} = (1 + u)L_0 + L_1 = (L_0 + L_1) + uL_0. \tag{5}$$

For $c = 1$, the lower block is zero because its entries lie in K_M and have zero trace upstairs; hence $L_0 = 0$ and

$$\lambda_{a,1}^{(2M)} = L_1 \in K_M. \tag{6}$$

Thus the closed dual basis simultaneously explains the scaled recursion and the unscaled descent. Equations (4) and the finite-field Artin–Schreier exact sequence are kernel-checked in `formal/Ogdoad/GoldDiagonal.lean`.

The source itself may also be constructed tower-recursively, with no final linear-algebra fallback. Write the quadratic extension as

$$K_{2M} = K_M(u), \qquad u^2 + u = \delta, \qquad \mathrm{Tr}_M(\delta) = 1.$$

The last equality is the Artin–Schreier irreducibility criterion. Given a trace-zero target $\mu = A+uB$, seek $w = X + uY$. Expanding gives

$$w^2 + w = (X^2 + X + \delta Y^2) + u(Y^2 + Y). \tag{7}$$

Because $\mathrm{Tr}_M(B) = \mathrm{Tr}_{2M}(\mu) = 0$, first solve $Y^2 + Y = B$ downstairs. The two choices are Y and $Y + 1$; replacing one by the other toggles the trace of $A + \delta Y^2$ by $\mathrm{Tr}_M(\delta) = 1$. Choose the one for which that trace vanishes, and solve

$$X^2 + X = A + \delta Y^2$$

downstairs. Recursing to K_1 constructs w throughout the Conway tower. The two final sources differ by 1; normalize by requiring the e_0 coordinate to vanish. At $a = 1$ this normalized source is exactly the Fermat-coin source of Proposition 2, giving

$$m = 8 : 20, \qquad m = 16 : 276, \qquad m = 32 : 65812.$$

The uniqueness of the normalized source makes this identification immediate from Proposition 2; the displayed tower recursion makes it constructive for every exponent. The case $a = 0$ is included algebraically, but $Q_0(x) = \mathrm{Tr}(x^2)$ is linear and has zero polar form, so it is not a genuinely quadratic Gold instance [20].

Remark 5 (subfield vanishing; standard math)**.** $q_i^{(m,a)} = 0$ for all $i < m/2$: the Fermat basis elements at lower indices, their Frobenius iterates, and their nim products stay inside the half-field, whose absolute trace from an even-degree extension vanishes. The Gold diagonal lives on the top Fermat layer only.

Remark 6 (diagonal framing)**.** The diagonal framing

$$q_i = Q_a(e_i) = \mathrm{Tr}(e_i^{1+2^a})$$

is game-built for every a: Proposition 2 gives the explicit $a = 1$ source, and Theorem 1 gives a constructive source for every exponent. The weighted-source Witt–FIFO theorem in Section 10 supplies a local normal-play rule with P-set $\{x : Q_a(x) = 0\}$.

7 The extraspecial reframing

The diagonal-framing question can be restated in group-theoretic terms, and the restatement sharpens it. The dictionary is classical – the characteristic-2 Heisenberg/extraspecial picture of quadratic forms [31, 21] – and this section proves the pieces we use, to keep the note self-contained. Claim levels: Lemmas 1, 2, 3 and Proposition 3 are standard mathematics; the R_8 instantiation in Corollary 1 is implemented and tested (`experiments/misere_kernel.py`); reading E-equivariance as a naturality criterion is interpretation. The weighted-source theorem below realizes the quadric without claiming that ordinary play itself is a model of the central extension E; that stricter extension-model question is optional.

Throughout, $V = \mathbb{F}_2^n$ written additively, and $Z = \{1, z\} \cong \mathbb{Z}/2$ written multiplicatively. A *quadratic map* $Q\colon V \to \mathbb{F}_2$ is a function with $Q(0) = 0$ whose polarization $B(x, y) = Q(x + y) + Q(x) + Q(y)$ is bilinear; B is then alternating (hence symmetric), and Q may have any diagonal $q_i = Q(e_i)$, exactly the char-2 datum the engine keeps separate from b.

7.1 Quadratic forms are central extensions

Lemma 1 (extraspecial dictionary). *Let $1 \to Z \to E \xrightarrow{\pi} V \to 1$ be a central extension of the elementary abelian 2-group V by $Z \cong \mathbb{Z}/2$.*

(i) For $x \in V$ and any lift $\tilde{x} \in \pi^{-1}(x)$ one has $\tilde{x}^2 \in Z$, and $\tilde{x}^2$ is independent of the lift; the squaring form $Q_E\colon V \to \mathbb{F}_2$ defined by $\tilde{x}^2 = z^{Q_E(x)}$ is well defined, as is the commutator pairing B_E defined by $[\tilde{x}, \tilde{y}] = z^{B_E(x,y)}$, which is bilinear and alternating. They satisfy

$$(\tilde{x}\tilde{y})^2 = \tilde{x}^2\, \tilde{y}^2\, [\tilde{x}, \tilde{y}], \qquad \text{equivalently} \qquad Q_E(x + y) = Q_E(x) + Q_E(y) + B_E(x, y),$$

so Q_E is a quadratic map with polar form B_E.

(ii) Conversely, every quadratic map Q arises: fixing a basis $e_1, \ldots, e_n$ of V, the bilinear 2-cocycle

$$\varphi(x, y) = \sum_i x_i y_i\, Q(e_i) + \sum_{i>j} x_i y_j\, B(e_i, e_j)$$

defines a group $E_Q = Z \times V$ with $(s, x)(t, y) = (s + t + \varphi(x, y),\, x + y)$, whose squaring form is Q and whose commutator pairing is the polar form B.

(iii) The assignment $Q \mapsto [E_Q]$ is a bijection from quadratic maps on V to equivalence classes of central extensions of V by $\mathbb{Z}/2$.

(iv) E_Q is abelian iff $B = 0$ (iff Q is linear); and for $n \geq 1$, E_Q is extraspecial (i.e. $Z(E) = [E, E] = \Phi(E) \cong \mathbb{Z}/2$) iff B is nondegenerate, which forces $n = 2r$ even and $|E_Q| = 2^{1+2r}$.

Proof. (i) $\pi(\tilde{x}^2) = 2x = 0$, so $\tilde{x}^2 \in Z$; replacing $\tilde{x}$ by $z\tilde{x}$ multiplies the square by $z^2 = 1$. Commutators of lifts lie in Z because V is abelian, are central, and are unchanged by central retagging of the lifts; centrality of commutators gives bimultiplicativity, and $[\tilde{x}, \tilde{x}] = 1$ gives $B_E(x, x) = 0$. For the identity: with $c = [\tilde{y}, \tilde{x}]$ central, $\tilde{x}\tilde{y}\tilde{x}\tilde{y} = \tilde{x}(\tilde{x}\tilde{y}\,c)\tilde{y} = \tilde{x}^2\tilde{y}^2 c$, and $c = c^{-1} = [\tilde{x}, \tilde{y}]$ since Z has exponent 2. As $\tilde{x}\tilde{y}$ lifts $x + y$, the additive identity follows.

(ii) A bilinear map is a 2-cocycle, so E_Q is a group with center containing $Z = \{(0, 0), (1, 0)\}$. Squares: $(s, x)^2 = (\varphi(x, x), 0)$ and $\varphi(x, x) = \sum_i x_i Q(e_i) + \sum_{i>j} x_i x_j B(e_i, e_j) = Q(x)$ by the polarization expansion of Q in the basis. Commutators: $(s, x)(t, y)(s, x)^{-1}(t, y)^{-1} = (\varphi(x, y) + \varphi(y, x), 0) = (B(x, y), 0)$, using that the symmetrization of φ is B (the diagonal terms cancel and B is symmetric).

(iii) Equivalent extensions have cohomologous cocycles, and a coboundary $\delta\psi(x, y) = \psi(x) + \psi(y) + \psi(x + y)$ has zero diagonal ($\delta\psi(x, x) = \psi(0) = 0$), so the squaring form is an invariant of the class; (ii) gives surjectivity. Injectivity: if two extensions have equal squaring form Q, choose sections and cocycles φ, φ'; both have diagonal Q, and both have symmetrization equal to the polar form of Q, so $d = \varphi + \varphi'$ is a symmetric cocycle with zero diagonal. The extension E_d is then abelian of exponent 2, i.e. an $\mathbb{F}_2$-vector space, so $1 \to Z \to E_d \to V \to 1$ splits and d is a coboundary; the two extensions are equivalent.

(iv) Commutators generate $z^{\operatorname{im} B}$, so E_Q is abelian iff $B = 0$. In general $Z(E_Q) = \pi^{-1}(\operatorname{rad} B)$, since $\tilde{x}$ is central iff $B(x, \cdot) = 0$; thus $Z(E_Q) = Z$ iff B is nondegenerate. If so, $B \neq 0$ gives $[E, E] = Z$, and $\Phi(E) = E^2[E, E] = Z$ because all squares lie in Z and E/Z is elementary abelian. A nondegenerate alternating form exists only in even dimension. □

Remark 7. This is the characteristic-2 Heisenberg picture: E_Q is the Heisenberg group of (V, B) and Q selects the central extension; cohomologically, (iii) is the standard identification of $H^2(V; \mathbb{F}_2)$ with quadratic maps $V \to \mathbb{F}_2$ [31]. The same finite-quadratic-module data drives the Weil S/T matrices in the library's integral layer. Note that the dictionary is exactly the engine's discipline in group clothing: the diagonal q_i (squares) and the off-diagonal b_{ij} (commutators) are independent central data, and collapsing them is collapsing E_Q onto an abelian quotient.

7.2 Arf classifies the two extraspecial types

Write H for the hyperbolic plane ($Q(e_1) = Q(e_2) = 0$, $B(e_1, e_2) = 1$; $\operatorname{Arf} = 0$) and A for the anisotropic plane ($Q \equiv 1$ on $V \setminus \{0\}$; $\operatorname{Arf} = 1$), and $\circ$ for the central product (identify the centers).

Lemma 2 (Arf = the two types)**.** *Let B be nondegenerate,* $\dim V = 2r \geq 2$.

(i) $E_{Q \perp Q'} \cong E_Q \circ E_{Q'}$.

(ii) $E_H \cong D_8$ *and* $E_A \cong Q_8$.

(iii) Every extraspecial 2-group of order 2^{1+2r} is isomorphic to E_Q for a nondegenerate Q, and $E_Q \cong E_{Q'}$ iff $(V, Q) \cong (V', Q')$ iff the ranks and Arf invariants agree [12, 2]. Hence there are exactly two extraspecial groups of order 2^{1+2r}:

$$E^+_{2^{1+2r}} = D_8^{\circ r} \qquad\qquad (\operatorname{Arf} = 0,\ Q \cong H^{\perp r}),$$
$$E^-_{2^{1+2r}} = D_8^{\circ(r-1)} \circ Q_8 \qquad\qquad (\operatorname{Arf} = 1,\ Q \cong H^{\perp(r-1)} \perp A).$$

In particular $D_8 \circ D_8 \cong Q_8 \circ Q_8$, the group avatar of $H \perp H \cong A \perp A$ (additivity of Arf*).*

(iv) (Census.) With $\varepsilon = (-1)^{\operatorname{Arf} Q}$,

$$\#\{g \in E_Q : g^2 = 1\} = 2\,\#\{x : Q(x) = 0\} = 2^{2r} + \varepsilon\, 2^r,$$

so the element-order census of E_Q is the zero-count bias of Section 4 read multiplicatively, and it distinguishes the two types.

Proof. (i) The cocycle $\varphi \oplus \varphi'$ on $V \oplus V'$ is bilinear with diagonal $Q \perp Q'$, and $(s, (x, x')) \mapsto [((s, x), (0, x'))]$ is an isomorphism onto $(E_Q \times E_{Q'})/\langle (z, z') \rangle$.

(ii) E_H has order 8, is nonabelian ($B \neq 0$), and has a noncentral involution (any lift of e_1 squares to $z^{Q(e_1)} = 1$), so $E_H \cong D_8$; in E_A every noncentral element squares to z (as $Q \equiv 1$ off 0), so z is the unique involution and $E_A \cong Q_8$.

(iii) For E extraspecial, $V = E/Z(E)$ is elementary abelian (as $\Phi(E) = Z(E)$), so Lemma 1 applies to $1 \to Z(E) \to E \to V \to 1$ and gives $E \cong E_{Q_E}$ with B_{Q_E} nondegenerate ($Z(E)$ exactly central). If $\psi\colon E_Q \to E_{Q'}$ is an isomorphism, it preserves the (characteristic) centers, fixes $z \mapsto z'$ (the unique nontrivial central element), and induces an $\mathbb{F}_2$-linear map $\bar{\psi}\colon V \to V'$ with $Q'(\bar{\psi}x)$ read from $\psi(\tilde{x})^2 = \psi(\tilde{x}^2)$ – an isometry. Conversely an isometry g transports the cocycle: $\varphi' \circ (g \times g)$ has diagonal $Q' \circ g = Q$, hence by Lemma 1(iii) yields an extension equivalent to E_Q, and $(s, x) \mapsto (s, gx)$ closes the isomorphism. Dickson's classification of nonsingular quadratic forms over $\mathbb{F}_2$ (two isometry classes per even rank, separated by Arf) finishes; the central-product normal forms follow from (i), (ii) and Witt decomposition.

(iv) The two lifts of x both square to $z^{Q(x)}$, so they are involutions or the identity iff $Q(x) = 0$; now apply the zero-count formula of Section 4. The counts differ for $\varepsilon = \pm 1$, so $E^+ \not\cong E^-$. □

7.3 The abelian obstruction

Lemma 3 (abelian obstruction)**.** *Let M be a commutative monoid, $Z = \{1, z\} \subseteq M$ a two-element subgroup, $N \subseteq M$ a submonoid containing Z, and $\pi\colon N \twoheadrightarrow V$ a surjective monoid homomorphism onto an $\mathbb{F}_2$-vector space with $\pi^{-1}(\pi(m)) = mZ$ for all $m \in N$. Then the squaring form $Q\colon V \to \mathbb{F}_2$, defined by $\tilde{m}^2 = z^{Q(x)}$ for any $\tilde{m} \in \pi^{-1}(x)$, is well defined and $\mathbb{F}_2$-linear; its polar form vanishes identically. Consequently no commutative monoid realizes, through its own squaring map, a quadratic form with nonzero – a fortiori nondegenerate – characteristic-2 polar form. Equivalently, by Lemma 1(iv): if $B \neq 0$, the extension E_Q is nonabelian and admits no model (N, Z, π) inside any commutative monoid.*

Proof. $\pi(m^2) = 2\pi(m) = 0$, so $m^2 \in \pi^{-1}(0) = Z$, and $(zm)^2 = z^2m^2 = m^2$, so Q is well defined. Commutativity gives $(mn)^2 = m^2n^2$; since $\tilde{m}\tilde{n}$ is a preimage of $x+y$, this is $Q(x+y) = Q(x)+Q(y)$, i.e. $B \equiv 0$. □

Corollary 1 (misère quotients: the kernel shadow is linear)**.** *Let $\mathcal{Q}$ be a finite misère quotient – a finite commutative monoid – with kernel K, the maximal subgroup of $\mathcal{Q}$ (the mutual-divisibility class of the product of all idempotents) [30].*

(i) Every configuration (N, Z, π) as in Lemma 3 inside $\mathcal{Q}$ has linear squaring form: the intrinsic multiplication of a misère quotient cannot supply a quadratic refinement with $B \neq 0$, on K or anywhere else in $\mathcal{Q}$.

(ii) Under the regularity hypothesis of [30, Thm. 6.4] (satisfied by the regular finite quotients arising in practice), $K \cong (\mathbb{Z}/2)^k$ is the group of normal-play Grundy values and the P-portion meets K in the XOR-linear normal-play set.

(iii) On the smallest wild quotient $R_8 = \langle a, b, c \mid a^2 = 1,\ b^3 = b,\ bc = ab,\ c^2 = b^2\rangle$ with $P = \{a, b^2\}$: the idempotents are $\{1, b^2\}$, the kernel is $K = \{b^2, b, ab, ab^2\} \cong (\mathbb{Z}/2)^2$ with identity b^2, $P \cap K = \{b^2\} \mapsto \{0\}$ is linear, and the genuinely misère P-element a lies outside K. (Implemented and tested: `experiments/misere_kernel.py`.)

So the linear obstruction observed on R_8 is forced by commutativity, not an accident of the example: a nondegenerate polar form is the commutator pairing of a nonabelian group, and a misère quotient has none to offer.

Proof. (i) is Lemma 3 applied to submonoids of $\mathcal{Q}$, which are commutative. (ii) is quoted from [30]. (iii) is a finite verification. □

7.4 The integral shadow

The same extension data has an integral-lattice avatar, already shipped in the library (interpretation, verified through the `DiscriminantForm` + `arf_nimber` pipeline): the Gold nonsingular core of rank $2r$ and Arf ε is the discriminant form of the even lattice $U(2)^r$ ($\varepsilon = 0$) or $U(2)^{r-1} \oplus D_4$ ($\varepsilon = 1$); the Milgram phase is $4 \cdot \mathrm{Arf} \mod 8$; the Weil S-matrix prefactor is $(-1)^{\mathrm{Arf}}$; and the Frobenius–Schur indicator of the Heisenberg representation of E_Q is $(-1)^{\mathrm{Arf}}$. The metaplectic relation $(ST)^3 = S^2$

holds iff the diagonal T uses a quadratic refinement of the correct Arf class – it selects the Arf *class*, not the member. So the integral bridge carries exactly one game-relevant bit; it cannot, by itself, distinguish the Gold form from any other refinement in its class.

7.5 An E-equivariance screen

Proposition 1 kills rules that are blind to everything but the symplectic structure (Tier 1), while per-position evaluator circuits for Q_a realize the quadric tautologically (Tier 3). The extraspecial dictionary suggests a criterion strictly between the two: the rule should see the extension E_Q – which carries the diagonal data q_i structurally, as squares – but only up to its automorphisms, so that no basis, field structure, or evaluation circuit can be smuggled in.

Definition 1 (uniform rules; the three tiers)**.** Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$ with polar form B, and $E = E_Q$, $\pi\colon E \to V$, $\Sigma = \{x \in V : Q(x) = 0\}$.

(a) A *uniform rule on a finite set* X is a single move relation $M \subseteq X \times X$, fixed once for all positions; for normal play we require the move digraph acyclic, and outcomes are computed by the usual retrograde recursion. (For loopy or misère readings, replace the outcome recursion; the equivariance constraint below applies verbatim, since outcome classes are invariants of digraph automorphisms.)

(b) A uniform rule on E is *E-equivariant* if every $\alpha \in \mathrm{Aut}(E)$ is an automorphism of the move digraph. Its *shadow* is $\pi(P) \subseteq V$, where P is its P-set; it *realizes* $T \subseteq V$ if $\pi(P) = T$.

(c) *Tier 1* ($\mathrm{Sp}(B)$-blind): a uniform rule on V invariant under $\mathrm{Sp}(B)$, as in Proposition 1. *Tier 3* (Q_a-evaluation): a family of games $\{\Gamma_x\}_{x\in V}$ whose structure varies with the designated input x (e.g. the trace-circuit evaluator); this is not a uniform rule. *Tier 2 screen*: a route to the Gold quadric is Tier-2 natural *only if* its positions and moves lift to an E-equivariant uniform rule on E_{Q_a} – with Q_a restricted to its nonsingular core when the form is degenerate, matching the classifier's radical discipline – that realizes $\{Q_a = 0\}$.

Proposition 3 (the screen sits strictly between the solved tiers)**.** *Let Q be nondegenerate on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, and $E = E_Q$.*

(i) The image of $\mathrm{Aut}(E) \to \mathrm{GL}(V)$ *is exactly the orthogonal group* $\mathrm{O}(Q)$*, with kernel the central twists $g \mapsto g\, z^{\ell(\pi g)}$, $\ell \in V^*$, which equal* $\mathrm{Inn}(E) \cong V$ *because B is nondegenerate; thus* $\mathrm{Out}(E) \cong \mathrm{O}(Q)$ *(cf. [37]). The* $\mathrm{Aut}(E)$*-orbits on E are*

$$\{1\}, \qquad \{z\}, \qquad \pi^{-1}(\Sigma \setminus \{0\}), \qquad \pi^{-1}(V \setminus \Sigma).$$

(ii) Hence the P-set of any E-equivariant uniform rule is a union of these four orbits, and its shadow is one of the eight unions of $\{0\}$, $\Sigma \setminus \{0\}$, $V \setminus \Sigma$*. The quadric Σ is among them: the Tier-1 exclusion does not apply.*

(iii) (Tier 1 $\subsetneq$ Tier 2.) Every $\mathrm{Sp}(B)$*-invariant uniform rule on V pulls back along π to an E-equivariant rule with P-set π^{-1} of the original, whose shadow is therefore one of $\varnothing$, $\{0\}$, $V \setminus \{0\}$, V – never the quadric (Proposition 1 in this language). The containment is strict: the squaring rule*

$$g \to h \textit{ legal} \iff g^2 = z \textit{ and } h^2 = 1$$

("move from any order-4 position to any position of order at most 2") is E-equivariant and acyclic, and its P-set is the involution locus $\{g : g^2 = 1\} = \pi^{-1}(\Sigma)$, with shadow exactly Σ.

(iv) *(Tier 2 $\subsetneq$ Tier 3.) Up to the isometry induced by any group isomorphism, the family of E-equivariant rules and their shadows depends only on $(r, \mathrm{Arf}\, Q)$ (Lemma 2(iii)). In particular the screen cannot separate Gold forms of equal core rank and Arf invariant, and an E-equivariant rule has no access to m, the exponent a, the field multiplication, the coordinate frame $q_i = Q_a(e_i)$, or any evaluation circuit. Conversely, arbitrary subsets of V – all realizable by ad hoc acyclic lookup games and by Tier-3 evaluators (Section 6.1) – are not shadows of E-equivariant rules once they leave the eight-set list of (ii).*

Proof. (i) Automorphisms preserve $Z = Z(E)$ and fix z ($\mathrm{Aut}(\mathbb{Z}/2) = 1$), so the induced map preserves the squaring form: $Q(\bar{\alpha}x)$ is read from $\alpha(\tilde{x})^2 = \alpha(\tilde{x}^2) = z^{Q(x)}$; the image lies in $\mathrm{O}(Q)$. Surjectivity: an isometry g transports the standard cocycle as in the proof of Lemma 2(iii), producing an automorphism of E_Q inducing g. The kernel consists of maps $g \mapsto g\, z^{\ell(\pi g)}$ with ℓ linear (multiplicativity forces additivity of ℓ); inner automorphisms are exactly the twists $\ell = B(\bar{g}, \cdot)$, and nondegeneracy makes $\bar{g} \mapsto B(\bar{g}, \cdot)$ onto V^*. Orbits: $\mathrm{O}(Q)$ is transitive on $\Sigma \setminus \{0\}$ and on $V \setminus \Sigma$ by Witt's extension theorem in its characteristic-2 quadratic-space form [35] (both sets are nonempty for $r \geq 2$); the two lifts of any $x \neq 0$ are conjugate under conjugation by $\tilde{g}$ with $B(\bar{g}, x) = 1$; and 1, z are fixed by everything.

(ii) Outcome recursion is invariant under digraph automorphisms. Thus P is $\mathrm{Aut}(E)$-invariant; apply (i) and project.

(iii) The pullback $M = \{(g, h) : (\pi g, \pi h) \in R\}$ of an $\mathrm{Sp}(B)$-invariant rule R is preserved by $\mathrm{Aut}(E)$, since the induced action on V lies in $\mathrm{O}(Q) \subseteq \mathrm{Sp}(B)$; it is acyclic when R is, and an easy induction along the acyclic rank gives $P(M) = \pi^{-1}(P(R))$. By the transitivity of $\mathrm{Sp}(B)$ on $V \setminus \{0\}$ for $r \geq 2$, $P(R)$ is a union of $\{0\}$ and $V \setminus \{0\}$. For the squaring rule: automorphisms preserve element orders, so the rule is E-equivariant; positions with $g^2 = 1$ are terminal, hence P; positions with $g^2 = z$ have a move (to 1), hence N. Its shadow Σ satisfies $1 < |\Sigma| = 2^{2r-1} + \varepsilon 2^{r-1} < 2^{2r} - 1$ for $r \geq 2$, so it is none of the four Tier-1 shadows.

(iv) If $\psi \colon E_Q \to E_{Q'}$ is an isomorphism (it exists iff ranks and Arf agree, Lemma 2(iii)), then $M \mapsto (\psi \times \psi)(M)$ is a bijection between E-equivariant rules carrying P-sets to P-sets and shadows to shadows through the induced isometry $\bar{\psi}$; in particular it carries rules realizing $\{Q = 0\}$ to rules realizing $\{Q' = 0\}$. The shadow constraint of (ii) excludes all but eight subsets, while Tier-3 constructions realize any subset; the containment is proper. $\square$

Remark 8 (what the screen does and does not settle)**.** Equivariance is a screen, not a construction. Over the *abstract* group E the screen is satisfiable – Proposition 3(iii) exhibits the squaring rule, which is nothing but the extraspecial squaring map of Lemma 1 read as a one-move relation. That rule consults only the multiplication of E, so it is a Q-evaluator exactly insofar as E itself is fed in by hand. The reframing replaces the request *find a play rule that computes Q_a* by the structural question *exhibit a game-native model of the extension E_{Q_a}* – positions built from game values, multiplication from game constructions – after which the rule layer is canonical. Lemma 3 is the sharp constraint on that search: no commutative value-world, in particular no misère-quotient kernel (Corollary 1), can host E once $B \neq 0$. The noncommutativity must enter from a structurally available asymmetry, and the one that normal, misère, and partizan play all possess – and that the symmetric polar form B discards – is the first-/second-player asymmetry of the move relation. Claim levels: the proposition is standard mathematics; treating E-equivariance as *the* naturality criterion is interpretation. The weighted-source rule resolves the P-set problem covariantly without first constructing the entire abstract extension as a game object; such a full model remains a stricter optional generalization. (For $r = 1$ the screen degenerates: on the anisotropic plane $\Sigma = \{0\}$ and $\mathrm{O}(Q) = \mathrm{Sp}(B)$, so the statement is vacuous there; the Gold targets of interest have $r \geq 2$ cores.)

Theorem E below sharpens the screen further: read as *invariance* it admits no live middle at all, so the E-structure must enter *covariantly*, in the dynamics.

Open question. *Does any game-native construction realize the extraspecial extension E_{Q_a} – equivalently, produce the diagonal data $q_i = Q_a(e_i)$ as squares in a noncommutative game-built structure – without an evaluation circuit for Q_a? By Lemma 3 the source cannot be any commutative game-value monoid.*

8 No-go theorems for the middle tier

The results below delimit commutative provenance, affine and orthogonal symmetry, bounded B-oracle access, B-local flip rules, and E-translation dynamics. Theorem A supplies the preceding coin-turning linearity result.

Theorem B (commutative squaring collapse; standard math)**.** *In any commutative monoid, squaring is an endomorphism, so any quadratic datum read through homomorphic coordinates has polarization identically zero. In particular the extraspecial squaring map Q (for $B \neq 0$) cannot live in, map into, or pull back through any commutative game monoid. This covers misère quotients, turn-parity counters, and all coin-turning configuration sums.*

Proof. Immediate from Lemma 3: commutativity gives $(mn)^2 = m^2n^2$, so the squaring form is additive and $B \equiv 0$. ☐

Conditional statement C (group misère quotients are tiny)**.** Assume [30, Thm. 6.4] with its regularity hypothesis. If the misère indistinguishability quotient $\mathcal{Q}(\mathcal{A})$ of a set of impartial games is a group, then $|\mathcal{Q}(\mathcal{A})| \leq 2$.

Proof sketch. Doubling lemma: in a group quotient every element is cancellable, and any position of Grundy value 2 doubles to a misère-P position, contradicting Grundy-distinctness from 0; the quoted structure theorem then pins the quotient to at most $\{1, a\}$. ☐

Remark 9 (scope and a probe caveat)**.** Conditional statement C depends on the Plambeck–Siegel structure theorem as cited; the regularity hypothesis is quoted from the repository's `experiments/misere_kernel.py` docstring, and the JCTA 2008 paper has *not* been independently re-verified within this program (closing that gate is a listed next move). Two consequences for the probe map: the $(\mathbb{Z}/2)^k$ ($k \geq 2$) target of `examples/octal_hunt.rs` is empty a priori – its empirical negatives (quotient orders $2, 6, 10, 12, 14$) are a consequence under the quoted hypothesis, not merely a sampling accident – and any future "hit" from that sweep would be a labeling artifact, since the helper `p_set_as_f2` checks only that the subset labeling is a bijection, not that it is a monoid homomorphism. The hunt's target should be reframed accordingly.

For the remaining results, fix a nondegenerate quadratic form Q on $V = \mathbb{F}_2^{2r}$, $r \geq 2$, with polar form B and zero set $\Sigma = \{Q = 0\}$, and write $\mathrm{AO}(Q) = \{x \mapsto gx + c : g \in \mathrm{Sp}(B),\ Q(gx + c) = Q(x)\ \forall x\}$ for the affine stabilizer appearing below.

Theorem D (affine symmetry budget)**.** $\mathrm{Stab}_{\mathrm{AGL}(V)}(\Sigma) = \mathrm{AO}(Q)$*; it contains no nontrivial pure translation (for a nonsingular core, $c \in \mathrm{rad}(B) = 0$), and its pure-linear part is exactly* $\mathrm{O}(Q)$*. Consequently every uniform rule whose move digraph admits even one affine automorphism outside* $\mathrm{AO}(Q)$ *has P-set $\neq \Sigma$.*

Proof. A Boolean-valued quadratic form is determined as a function by its zero set. Thus an affine map stabilizes Σ exactly when it preserves Q, which after polarization forces its linear part to preserve B. This proves $\mathrm{Stab}_{\mathrm{AGL}(V)}(\Sigma) = \mathrm{AO}(Q)$. If the pure translation $x \mapsto x + c$ preserves Q, evaluation at $x = 0$ gives $Q(c) = 0$, and then

$$0 = Q(x + c) + Q(x) = B(x, c) \qquad \text{for every } x.$$

Hence $c \in \mathrm{rad}(B) = 0$. With $c = 0$, preserving Q is precisely membership in $\mathrm{O}(Q)$. $\square$

Theorem E (the symmetry dial has no middle)**.** *The classical Dye–Dieudonné maximality theorem gives that* $\mathrm{O}(Q)$ *is maximal in* $\mathrm{Sp}(B)$ *[15, 13], and the* $\mathrm{O}(Q)$*-orbitals on* $V \times V$ *are exactly the Gram classes* $(Q(u), Q(w), B(u, w))$ *together with the degeneracy flags (Witt extension; see [16, §8]). Hence a rule invariant under any group strictly larger than* $\mathrm{O}(Q)$ *is dead by Theorem D (its symmetry reaches outside* $\mathrm{AO}(Q)$*), while a rule fully* $\mathrm{O}(Q)$*- or* $\mathrm{Aut}(E)$*-equivariant factors extensionally through pointwise* Q*- and* B*-evaluation. There is no invariance class strictly between dead (too coarse) and Gram-factoring (an evaluator):* E*-naturality must be covariance, not invariance.*

Corroboration. At $m = 4$, every one of the 648 (resp. 600) elements of $\mathrm{Sp}(4, 2)$ outside $\mathrm{O}^+(4, 2)$ (resp. $\mathrm{O}^-(4, 2)$) generates $\mathrm{Sp}(4, 2)$ together with the orthogonal group; orbitals enumerated exhaustively at $m = 4$ (`experiments/gold/ao_orbitals.py`).

Theorem F (B-oracle lower bound)**.** *Access model: legality of a move is decided from XOR combinations of the two positions and* t *fixed constant vectors, with oracle access to* B *only. If* $t \leq 2r - 3$*, then for every quadratic refinement* Q' *of* B *simultaneously there is a nonzero* Q'*-singular vector in the orthogonal complement of the constant span; the transvection it induces is a digraph automorphism lying outside* $\mathrm{O}(Q')$*, and Theorem D kills the* P*-set. The bound is tight: at* $t = 2r - 2$ *with anisotropic complement the argument fails. Hence the Gold diagonal framing* $q_i = Q_a(e_i)$*, which supplies* $2r$ *frame constants, is within two dimensions of information-theoretically forced.*

Proof. Let W be the span of the t constants. Since $\dim W^\perp \geq 3$, the restriction of Q' to $W^\perp$ has a nonzero singular vector v (a totally anisotropic binary quadratic space has dimension at most two). The symplectic transvection $T_v x = x + B(x, v)v$ fixes every constant. Writing $b = B(x, v)$ and using $Q'(v) = 0$ gives

$$Q'(T_v x) = Q'(x) + bQ'(v) + bB(x, v) = Q'(x) + B(x, v).$$

Thus T_v preserves B and every allowed oracle expression but does not preserve Q'; nonsingularity of B supplies an x with $B(x, v) = 1$. Theorem D applies. When $\dim W^\perp = 2$, its restriction may be anisotropic, explaining the stated sharp boundary of this argument. $\square$

Corroboration. The exact escape boundary $t = 2r - 2$ was machine-checked at $m = 4$; an end-to-end spot check retrograde-solved 50 random $t = 1$ B-oracle rules (`experiments/gold/skeptic_nogo_check.py`, `nogo_verify.py`).

Theorem G (B-local flip rules)**.** *Any rule of the form "flip* d *at* v *iff* $f(d, B(v, d))$*" is automatically undirected, because* B *is alternating:* $f(d, B(v + d, d)) = f(d, B(v, d) + B(d, d)) = f(d, B(v, d))$*. For undirected loopy games the outcome partition collapses:* $\mathrm{Win} = \varnothing$, $\mathrm{Loss} =$ *the isolated vertices* $=$ *an affine flat, and* $\mathrm{Loss} \cup \mathrm{Draw} \neq \Sigma$ *for* $r \geq 2$ *(the count* $|\Sigma| = 2^{r-1}(2^r + \varepsilon)$ *has an odd factor* > 1*). This covers the Loss, Draw, and* $\mathrm{Loss} \cup \mathrm{Draw}$ *targets at once. The symmetric-*B *rule of* `examples/loopy_quadric.rs` *is the special case* $f(d, \beta) = \beta$*; its* $(4, 1)$ *"hit" is the radical coincidence of Section 6.1.*

Verification. Undirectedness is the two-line identity above (standard math); the loopy outcome collapse including the Draw-set gap is machine-checked at $m = 4$ (`experiments/gold/skeptic_nogo_check.py`).

Theorem H (E-translation no-go)**.** *For $r \geq 2$, no left-E-equivariant rule on $E = E_Q$ with moves $g \to gn$, $n \in N$ for a fixed nonempty $N \subseteq E$, has P-set equal to the involution locus $I = \{g : g^2 = 1\} = \pi^{-1}(\Sigma)$.*

Proof. $I \cdot I = E$: for every $v \neq 0$, the count $\#\{x : Q(x) = 0 = Q(x + v)\} = 2^{2r-2} + (-1)^{\mathrm{Arf}\,Q} 2^{r-1}$ is positive for $r \geq 2$ (character sum), so every $n \in N$ factors as $n = gh$ with $g, h \in I$. Then $g \in I$, and $g \cdot n = g(gh) = g^2 h = h \in I$: the move $g \to gn$ is an edge from I to I. A normal-play P-set admits no P-to-P edge, so $P \neq I$. □

Verification. Cross-checked on the bent $(8, 1, \lambda = 2)$ extraspecial group (`experiments/gold/extraspecial_core.py`).

8.1 Normal-play rigidity and the synthesis

The strongest constraint is not any single architecture kill but a rigidity statement inside a declared access class. For this subsection, call a rule *in-class* only if it is a *single-board vector-position rule*: its positions are exactly the vectors of V. In addition, the quadratic refinement is quarantined behind a weight-bounded oracle (the (w_0, c) metering of Definition 2 below), and the rule is *refinement-uniform*: it realizes the target for every refinement q of B in the declared family, not just the Gold one.

Theorem 2 (normal-play rigidity, in-class)**.** *Under the bounded-framing oracle model with refinement uniformity, every legal edge between vector positions of Hamming weight $> c\,w_0$ must flip Q – in normal, misère, and loopy-Loss semantics alike. Consequently every in-class realizer of Σ in those semantics is a Q-alternating ender: play telescopes Q, and the outcome is the parity of $Q(x)$.*

Proof sketch. The case $Q(v) = Q(w) = 0$ of a non-flipping edge is forbidden directly by the outcome structure in all three readings. The case $Q(v) = Q(w) = 1$ is reduced to it by an adversary substitution: replace q by $q' = q + \ell$ with ℓ a linear form annihilating every queried functional but not the dense indicators; refinement uniformity transports the rule to q', where the edge becomes a forbidden 0-to-0 edge. □

Theorem 3 (no live middle, relative to the quarantine)**.** *Let Q have nonsingular core of rank $2r \geq 4$. No fixed uniform single-board vector-position rule satisfying*

- *(S) no affine digraph automorphism outside* $\mathrm{AO}(Q)$*,*
- *(A) legality from B-oracle queries at $t \leq 2r - 3$ frame constants,*
- *(C) no commutative squaring route for the provenance of Q,*
- *(E) no left-E-equivariant structure on the extraspecial group,*
- *(R) bounded-framing oracle access with refinement uniformity,*

has P/Loss/Draw target equal to Σ with any bulk strategic content: under (R), every legal edge between dense positions flips Q, all non-alternating strategic content is confined to a Hamming ball of radius $c\,w_0$.

Remark 10 (scope of the no-go theorem)**.** Theorem 3 is relative to its hypotheses. It does not cover:

1. *Loopy-Draw semantics*: the substitution contradiction of Theorem 2 does not fire (Draw-to-Draw edges are legal).
2. $t \geq 2r - 2$ *with anisotropic complement*: escapes Theorem F at the symmetry level.
3. *Frobenius-aware access*: the Galois group lies inside $\mathrm{O}(Q_a)$, so all symmetry methods are silent; Theorem F's model explicitly excludes Frobenius-twisted queries.
4. *State-extended rules*: the weighted-source construction has FIFO positions $(U, \mathcal{Q}, \mathrm{ko}, \sigma, \phi)$ rather than positions in V. This is its escape from the single-board hypothesis; it still satisfies N1 uniformly over every refinement, so game-built Gold sourcing is not an escape from torsor uniformity.
5. *Rank* $r = 1$ *and radical-anisotropic degenerate layers*: the character-sum count in Theorem H vanishes, and small extraspecial groups can be realized.

The main construction uses the fourth route: a public Witt-frame lift with a genuinely larger FIFO state space. Section 9 replaces vocabulary quarantine by the invariant transcript-span boundary, and Section 10 attains that boundary with an explicit local rule. The list therefore records the theorem's scope, not unresolved premises of the Gold realization.

9 Local realization and the observation boundary

Two natural first attempts at "naturality" fail for structural reasons. *Equivariance fails*: by Theorem E there is no invariance class strictly between symmetry groups that kill the P-set and symmetry groups that force extensional factoring through Q- and B-evaluation. *Vocabulary quarantine fails*: the Gold diagonal $q_i^{(m,1)} = \mathrm{Tr}(e_i^3)$ is expressible as $\mathrm{Tr}(\wp(w)e_i)$ using only XOR, nim-product, and trace (Proposition 2) – the licensed operation chain itself – so no principled line separates "computing q" from "computing Q" on vocabulary alone. The criterion below replaces both with access metering and a sharp transcript-observation boundary.

Definition 2 (uniform local realization)**.** Fix the coin frame $\{e_i\}$ (the Grundy basis $g(n) = 2^n$, game-natural). *Public data*: the frame, $\oplus$, nim-product, Frobenius, and the full alternating form B (game-built via Turning Corners, Frobenius, and trace). *Quarantined data*: the quadratic refinement, presented as a weight-bounded oracle $O_q(z) = Q_q(z)$ for $\mathrm{wt}(z) \leq w_0$. A *uniform local rule* satisfies:

F1 (*q-blind statics*) position set, loading map, and terminal set are computed without oracle access;

F2 (*metered access*) each move predicate and state transition together use at most c adaptive oracle queries, each at a point of Hamming weight at most w_0, with (w_0, c) constants independent of m;

F3 (*declared semantics*) one canonical outcome labeling (normal, misère, loopy, or q-blind terminal payoff).

It is an *exact local realizer* if additionally:

N1 (*torsor uniformity*) for every m, every B in the declared family, and every refinement q of B: $\mathrm{outcome}(\iota(x)) = \mathrm{target}$ iff $Q_q(x) = 0$;

N2 (*locality budget*) the (w_0, c) metering of F2;

These are objective properties of a declared rule and oracle interface. They do not, by themselves, turn the informal adjective "natural" into a semantic invariant.

Theorem 4 (Tier-3 exclusion under N1+N2)**.** *For any position x with $\mathrm{wt}(x) > c\,w_0$, the at most c queried functionals span only weight-$\le c\,w_0$ directions, so there exist refinements q, q' agreeing with every oracle answer but differing on $Q(x)$. Hence no in-class rule's move predicate factors through $Q(x)$ at dense positions – by theorem, not stipulation.*

Definition 3 (transcript stability)**.** Let $E(q, z)$ be the evaluation returned by refinement oracle q, let $S(q, x)$ be the finite set of observations selected at (q, x), and let $R(q, x)$ be the rooted result. The triple (E, S, R) is *transcript-stable* if, for every q, q' and x,

$$\big(E(q, z) = E(q', z) \text{ for every } z \in S(q, x)\big) \implies R(q, x) = R(q', x).$$

The observation set may depend adaptively on the original oracle q; the definition says that agreement on the resulting transcript fixes the result. This is the definition formalized in `formal/Ogdoad/GoldNoEvaluator.lean`.

Definition 4 (bounded outcome certificate)**.** Write $\mathrm{N2cert}(C, w)$ when, for every refinement q and input x, there is a finite observation set $S(q, x)$ with $|S(q, x)| \le C$ and $\mathrm{wt}(z) \le w$ for $z \in S(q, x)$ such that every same-polar refinement q' agreeing with q on $S(q, x)$ has the same rooted outcome. The set may be chosen adaptively from q; what is bounded is the complete certificate for the outcome, not the number of queries made by one transition.

The algebra behind the next bound is standard: quadratic Boolean functions with a fixed polar form differ by linear functions, and agreement on queried linear directions is controlled by the annihilator of their span, the usual linear-structures viewpoint [7]. Linear queries are likewise the primitive of parity decision trees [39]. The novel point here is transporting that characterization to an adaptive game transcript through Definition 3.

Theorem 5 (transcript-span observation lower bound)**.** *Let the refinement family at fixed B be closed under addition of every linear functional. If a transcript-stable rooted result is exact at x, then*

$$x \in \operatorname{span} S(q, x).$$

In the original frame, if every observed vector has weight at most w, then

$$\mathrm{wt}(x) \le |S(q, x)|w.$$

Consequently no constants C, w give an exact torsor-uniform family satisfying $\mathrm{N2cert}(C, w)$ *in unbounded dimension.*

Proof. If $x \notin \operatorname{span} S$, finite-dimensional separation gives a linear functional ℓ vanishing on S but not at x. The refinements q and $q + \ell$ therefore have identical observation transcripts and rooted outcomes, while their target values at x differ, contradicting exactness. For the coordinate bound, the union of the supports of $|S|$ weight-w vectors has size at most $|S|w$ and must cover the support of x. Both forms are kernel-checked in `formal/Ogdoad/GoldNoEvaluator.lean`. □

At weight $w = 1$, Proposition 4 attains the bound exactly with the singleton support of x. For $w > 1$ the theorem is only a lower bound; no matching construction is claimed.

Theorem 6 (fork padding obstruction)**.** *The properties "there exists a refinement-sensitive winning fork at a reachable position" and the stronger variant requiring that fork after every original completed (hence every optimal completed) play do not certify that a finite normal-play realizer is non-evaluative. Every finite normal-play tree can be padded, without changing its root outcome, so that every completed play enters an always-N two-action fork whose unique winning action swaps with one chosen refinement bit.*

Proof. Replace every terminal P-node by a one-option wrapper whose child is the always-N swapping fork. The wrapper is still P, so induction over the finite tree preserves every ancestor outcome. The new fork is forced after every original completed play. The construction and induction are kernel-checked as `win_padTerminals` in `formal/Ogdoad/GoldForkPadding.lean`. $\square$

Remark 11 (Why the old N3 screens are retired)**.** The old mixed-successor N3 is defeated by dominated escape edges, while after dominance pruning every retained successor at a position has one outcome class. Counterfactual winning-action variants remain vulnerable to Theorem 6, even if their witness is required on every optimal play. Thus the paper uses the representation-invariant certificate lower bound of Theorem 5 for the negative boundary and states the positive rule's syntax and access discipline directly. Strategic liveness remains useful descriptive evidence, not an axiom defining naturality.

10 The weighted-source Witt–FIFO construction

The arbitrary-graph isolated-dummy linking conjecture is studied separately as an open problem in `writeups/linking_affine.tex`; it is stronger than the present realization requires. The polar form is public data, so it may be put into a deterministic Witt frame before play. This changes the strategic support graph from an arbitrary induced graph to a matching. A naive version would attach the dense value $Q(f_i)$ to a Witt-basis coin f_i, violating N2 in the original Grundy frame. The decisive point is to split that value into a public polar part and weight-one source *interactions*: the refinement bits weight local overlap edges rather than being accumulated as invariant terminal labels.

Fix the original frame $e_1, \dots, e_m$ and write

$$P_B(z) = \sum_{j<k} B(e_j, e_k) z_j z_k, \qquad Q_q(z) = P_B(z) + \sum_j q_j z_j, \qquad q_j = Q_q(e_j).$$

Choose, by a fixed Gaussian-elimination convention depending only on B, a symplectic-plus-radical basis

$$f_i = \sum_j C_{ji} e_j, \qquad B(f_{a_h}, f_{b_h}) = 1$$

whose other pairings vanish. For $x = \sum_i y_i f_i$, put

$$p_i = P_B(f_i) = \sum_{j<k} C_{ji} C_{ki} B(e_j, e_k).$$

All of C, $y = C^{-1}x$, and p_i are refinement-blind public data.

For completeness, the adapted basis exists by the usual two-line induction for alternating forms. If $B = 0$, retain any basis as radical vectors. Otherwise choose the lexicographically first pair u, v with $B(u, v) = 1$. The plane $H = \langle u, v \rangle$ is nonsingular, since a vector $au + bv$ orthogonal to both u and v has $a = b = 0$; hence $V = H \oplus H^{\perp}$. Recurse on $H^{\perp}$. Choosing the first available pivot and performing row operations in the fixed original-coordinate order makes the construction deterministic. It uses only the public matrix B, works unchanged in the singular case, and terminates after one dimension drop for a radical step or two for a hyperbolic-plane step.

Rule (weighted-source WITT-FIFO). The public state is $(U, \mathcal{Q}, \mathrm{ko}, \sigma, \phi)$: the untouched coin set, a FIFO queue of open coins, a Boolean ko bit, the accumulated charge, and the mover-parity bit. Initially $\mathcal{Q}$ is empty, $\mathrm{ko} = \text{false}$, and $\sigma = \phi = 0$.

Load a strategic coin F_i for each $y_i = 1$, with fixed public diagonal charge p_i. Join F_{a_h} to F_{b_h} exactly when both are loaded. Every loaded strategic Witt edge has weight one. For each original coordinate $x_j = 1$, load in addition a q-blind source pair S_j^0, S_j^1. Its potential edge has local weight $q_j = O_q(e_j)$: opening one endpoint while its mate is already queued xors q_j into the charge. Source closes have charge zero.

Every loaded coin must be touched twice, first by OPEN and then by CLOSE. OPEN(v) is legal exactly when $v \in U$; it removes v from U, appends it to $\mathcal{Q}$, and sets ko to true exactly when the old queue was empty. Thus ko protects the newly opened front by forbidding CLOSE for one move. A subsequent OPEN clears ko because its old queue is nonempty. CLOSE is legal exactly when $\mathcal{Q}$ is nonempty and ko is false; it removes the FIFO front. PASS is forced exactly when $U = \varnothing$, $\mathcal{Q} \neq \varnothing$, and ko is true; it only clears ko. These are all FIFO moves.

When a coin v opens, xor into σ the weight of its already-open mate, if any. Thus every strategic or source edge is charged exactly when its two touch intervals overlap. When a strategic front F_i closes, update

$$\sigma \oplus= p_i.$$

All other closes and all passes leave σ unchanged. Toggle ϕ after every FIFO move. Each FIFO move strictly decreases

$$4|U| + 2|\mathcal{Q}| + [\mathrm{ko}],$$

so play is finite. Once $U = \varnothing = \mathcal{Q}$, add one ordinary normal-play claim move exactly when $\sigma = 1 \oplus \phi$. Taking that move ends the game at an optionless position; if it is absent, the drained state is already terminal.

For a Gold form, Theorem 1 supplies $q_j = \mathrm{Tr}((w_a^2 + w_a)e_j)$ for the source-pair weight. Definition 2 counts this as one weight-one oracle observation on an overlapping source OPEN. It does not claim that running the tower recursion for w_a, or evaluating its trace, is itself one primitive game move.

The move-parity bit is not an evaluator or form oracle; it is already the mover field of the FIFO state. It is necessary for an identity-valued payoff compiler: a phase-free terminal leaf assigns opposite initial-seat winners at even and odd depths. The final claim is therefore a two-state local "charge lantern", not an inspection of x, B, or $Q(x)$.

Theorem 7 (FIFO matching theorem)**.** *Let every component of G be K_1 or K_2. From the empty-queue root, either designated seat has a strategy forcing the sum of all close charges $\deg_U(f)$ to be zero, where*

$$\deg_U(f) := |\{u \in U : \{f, u\} \in E(G)\}| \pmod 2$$

is the live degree of the FIFO front. In fact the strategy makes every close charge zero individually. The theorem holds for every finite board and needs no dummy.

Proof. Call a FIFO front *safe* when either ko is set or its unique mate is not untouched; equivalently, every legal close at that checkpoint has charge zero. Induct on $4|U| + 2|\mathcal{Q}| + [\mathrm{ko}]$. We show that a designated seat forces terminal score zero whenever it moves from score zero, and whenever its opponent receives a score-zero safe state.

If the designated seat moves with $U \neq \varnothing$ and the queue is empty, OPEN any $z \in U$; ko makes the next checkpoint safe. If the front f has charge one, its unique mate u lies in U; OPEN u, which erases the only possible live neighbour of f. If f has charge zero, OPEN any $z \in U$; maximum degree one implies that erasure preserves the zero front. When $U = \varnothing$, all later closes have charge zero and the forced PASS, when present, is harmless.

At an opponent checkpoint, OPEN and PASS preserve the score and return the turn. A legal CLOSE rules out ko, so safety makes its charge zero and likewise returns a score-zero state. Every branch has smaller rank. Initially the queue is empty: the second seat receives a vacuously safe root, while the first seat uses the empty-queue OPEN. The designated seat itself never closes while an untouched coin remains, and after the untouched set is empty every close has charge zero. Thus the constructed strategy makes each close charge zero, not only their total. This proves both seats. $\square$

The exact transition-system proof is kernel-checked as `evenWins_initial_of_matching` in `formal/Ogdoad/FifoMatching.lean`. The finite matching census helped find the policy but is not used in the theorem. Its refinement-blind strengthening, with a public potential matching and an arbitrary edge-deleted scoring submatching, is kernel-checked as `evenWins_initial_of_every_submatching`. The proposition is pointwise in the scoring submatching, while its proof branches choose moves only from the public matching. Since `EvenWins` does not reify a first-class policy or complete play traces, the literal single-policy quantifier and the per-close-zero observation are the explicit argument above, not separate theorem statements in Lean. This per-close-zero strengthening is the one load-bearing step of the exactness proof that is not itself kernel-checked; Section 11 records it as the next Lean follow-up.

A *charge stance* is the choice of which initial seat is designated by Boolean charge 1 (equivalently, which charge that seat seeks in the payoff tree). The phase-aware claim compiler works for either choice.

Theorem 8 (weighted-source exactness)**.** *For every $\mathbb{F}_2$-valued quadratic refinement Q_q on a finite-dimensional $\mathbb{F}_2$-space, every input x, and either charge stance, weighted-source* WITT-FIFO *forces terminal charge $\sigma = Q_q(x)$. Consequently its charge-lantern normal-play root is P if and only if $Q_q(x) = 0$.*

Proof. Quadratic expansion in the adapted basis gives

$$P_B(x) = \sum_i y_i P_B(f_i) + \sum_{i<k} y_i y_k B(f_i, f_k) = \sum_i y_i p_i + \sum_h y_{a_h} y_{b_h}.$$

The linear refinement part transports without any dense query:

$$\sum_i y_i \sum_j C_{ji} q_j = \sum_j x_j q_j.$$

Therefore

$$Q_q(x) = \sum_i y_i p_i + \sum_j x_j q_j + \sum_h y_{a_h} y_{b_h}. \tag{8}$$

Form instead the q-blind *potential matching* P: retain every loaded strategic Witt edge and every loaded source-pair edge, irrespective of its weight q_j. Apply the explicit safe-front strategy of Theorem 7 to P. The policy consults only this potential matching and the public FIFO state: when the front's potential mate is still untouched it opens that mate, and otherwise it opens any untouched coin. It is therefore the same strategy for every refinement q. The proof of the theorem shows more than cancellation of a total score: every close has zero live P-degree.

For one potential edge e, exactly one of the following occurs: its two touch intervals overlap, or the earlier-opened endpoint closes while the other endpoint is still untouched. The latter case would give that close live P-degree one, so the strategy excludes it. Consequently every potential pair overlaps. The rule's total OPEN charge is therefore directly

$$\sum_h y_{a_h} y_{b_h} + \sum_j x_j q_j,$$

with no refinement-dependent auxiliary graph or strategy. Every active strategic F_i closes exactly once and supplies $\sum_i y_i p_i$. Equation (8) therefore gives forced terminal charge $Q_q(x)$ at both stances.

For the final assertion, retain the finite charge game tree. At a leaf of charge σ and phase ϕ, the unique claim exists exactly when the player to move is the seat designated by σ. Backward induction says that the compiled normal-play winner equals the charge-game winner at every subtree and phase. At the root, seat zero plays for charge one, so the root is N for $Q_q(x) = 1$ and P for $Q_q(x) = 0$. The general compiler and root equivalence are kernel-checked in `formal/Ogdoad/GoldSemantics.lean`. □

Proposition 4 (locality and optimal observation support)**.** *The rule satisfies F1–F3 and N1–N2 with $(w_0, c) = (1, 1)$. Its distinct refinement observations at input x are exactly the singleton directions $\{e_j : x_j = 1\}$; they span x and attain the weight-one lower bound of Theorem 5.*

Proof. The basis algorithm, $x \mapsto y$, both loaded support sets, the matching potential edges, public charges p_i, and terminal board condition use only B, x and the fixed frame. Thus F1 holds. An OPEN involving a source pair needs at most the one singleton value $O_q(e_j)$, and only when its mate is already queued; every other move predicate and transition is refinement-blind. Even the forcing strategy is refinement-blind: it is the safe-front policy on the public potential matching P. This proves F2 and N2 with $(1, 1)$. The semantics is the single normal-play claim rule, giving F3, and Theorem 8 is N1 for every refinement of B.

There is one source pair exactly when $x_j = 1$, so the set of possible oracle directions is exactly $\{e_j : x_j = 1\}$. These vectors are independent and sum to x; no proper subset spans x. Theorem 5 says every transcript-stable exact weight-one interface must cover the same support. □

The oracle notation is accounting, not hidden information: q is a public parameter of the perfect-information game. The play-dependent source-pair repair is essential. A rule using only invariant source charges would make every complete history have the form

$$\sigma_q(h) = c_{B,x}(h) + \sum_{j : x_j = 1} q_j,$$

so two refinements agreeing on $Q(x)$ produced action-labelled isomorphic game trees. In the present rule the coefficient of q_j is the play-dependent overlap of S_j^0, S_j^1; refinement data therefore enters the same local interaction law as the polar edges, not a history-independent terminal bit. This is an intensional design fact, not an outcome-only certification: Theorem 6 proves that the reachable and unavoidable fork properties do not provide such a certification.

The rank-zero boundary is left undecorated. The same weighted-source rule still realizes the linear refinement exactly; there simply are no strategic Witt edges to pretend otherwise.

The original isolated-dummy arbitrary-graph linking conjecture remains an interesting strict generalization, but it is no longer load-bearing for Gold: the public Witt frame reduces precisely the required boards to the proved matching class, while weighted source pairs preserve fixed-frame locality. Its proof frontier and affine-response reductions belong to the separate open-problem paper `writeups/linking_affine.tex`.

11 Formal boundary and extensions

The headline realization and observation bound separate into five independently checked components; a sixth module checks the auxiliary padding obstruction:

- `formal/Ogdoad/GoldDiagonal.lean`: trace blocks and the Artin–Schreier identities behind the all-exponent source;
- `formal/Ogdoad/FifoMatching.lean`: the both-seat matching strategy;
- `formal/Ogdoad/GoldMatchingAlgebra.lean`: the adapted-basis expansion;
- `formal/Ogdoad/GoldSemantics.lean`: the phase-aware claim compiler;
- `formal/Ogdoad/GoldNoEvaluator.lean`: the observation lower bound;
- `formal/Ogdoad/GoldForkPadding.lean`: outcome-preserving padding.

The paper supplies the concrete Witt basis, the original-frame coefficient transport, and the synthesis of those components. There is no single end-to-end Lean theorem constructing the complete arena. In particular, the paper's strategy-relative statement that one public policy makes every potential close degree zero is stronger than the pointwise submatching theorem presently encoded in Lean. Formalizing that single-policy, per-close-zero strengthening is the explicit follow-up; it is the only non-kernel-checked step used by the exactness synthesis.

The arbitrary-graph isolated-dummy FIFO theorem remains open and is studied in `writeups/linking_affine.tex`. It would strengthen FIFO combinatorics but does not strengthen Theorem 8, whose boards are matchings plus isolates. The all-exponent diagonal source is supplied by Theorem 1; it is not an additional hypothesis.

12 Brown phases as intrinsic partizan outcomes

Let V be a finite $\mathbb{F}_2$-space and let

$$q(x+y) = q(x) + q(y) + 2b(x,y), \qquad q\colon V \longrightarrow \mathbb{Z}/4,$$

where b is symmetric bilinear. Write

$$G(q) = \sum_{x\in V} i^{q(x)}.$$

Whenever $G(q) \neq 0$, define $\beta(q) \in \mathbb{Z}/8$ by $G(q)/|G(q)| = \exp(\pi i\beta(q)/4)$. This definition includes singular b when its Gauss sum is nonzero; when the sum vanishes there is no phase, and the radical is retained as separate data.

The next result is a convenient coordinate form of standard Brown-form material: Brown's relation $2Q(u) = jB(u,u)$, Wood's classification, and the same two-digit decomposition in the generalized-bent literature [6, 38, 32]. The correlated Walsh formula is the $q = 4$ case of the 2-adic generalized-bent decomposition [34, 33].

Theorem 9 (canonical Brown split)**.** *There are unique maps $\ell\colon V \to \mathbb{F}_2$ and $Q\colon V \to \mathbb{F}_2$ such that*

$$q = \widehat{\ell} + 2Q, \qquad B_Q = b + \ell \otimes \ell,$$

where $\widehat{0} = 0$ and $\widehat{1} = 1$ in $\mathbb{Z}/4$. Here ℓ is linear and Q is an ordinary characteristic-2 quadratic form with alternating polar form B_Q. Conversely, these data recover q and b. Moreover, with $W(R) = \sum_x (-1)^{R(x)}$,

$$G(q) = \frac{1+i}{2} W(Q) + \frac{1-i}{2} W(Q+\ell). \tag{9}$$

Proof. Reduction modulo 2 makes $\ell = q \bmod 2$ linear, and the two binary digits of $q(x)$ define $Q(x)$. Comparing the two expressions for $q(x+y)$, using $\widehat{a+c} = \widehat{a} + \widehat{c} + 2ac$, gives $B_Q = b + \ell \otimes \ell$. Finally, $i^\ell = (1+i)/2 + (1-i)(-1)^\ell/2$, which gives (9). □

Pointwise, the two coordinates and the Brown weight are

$q(x)$	0	1	2	3
$\ell(x)$	0	1	0	1
$Q(x)$	0	0	1	1
$i^{q(x)}$	1	i	-1	$-i$.

Thus (9) is a correlated, rather than factorized, pair of quadratic censuses. In particular, the Brown phase is not recovered from a marginal count for ℓ and an independent Arf count for Q.

Remark 12 (singular boundary)**.** Let $D = \operatorname{rad} b$. The diagonal identity $b(r,r) = q(r) \bmod 2$ gives $\ell|_D = 0$ and $q|_D = 2Q|_D$. If some $r \in D$ has $Q(r) = 1$, translation by r reverses every summand of $G(q)$, so $G(q) = 0$. If $Q|_D = 0$, the pair (ℓ, Q) descends to V/D, and the definition of β above is exactly the phase on that nonsingular quotient. This is why a zero Gauss sum is not assigned an artificial Brown octant.

Proposition 5 (Brown phase in quadratic coordinates)**.** *Suppose that b is nondegenerate. This is a re-coordinatization of the classification formulas of Wood [38].*

1. *If $\dim V$ is even, then B_Q is nondegenerate. If $\ell(x) = B_Q(a, x)$, then*

$$\beta(q) = 4\operatorname{Arf}(Q) + 2Q(a) \pmod 8.$$

2. *If $\dim V$ is odd, then $\operatorname{rad} B_Q = \langle w \rangle$, $\ell(w) = 1$, and B_Q is nondegenerate on $H = \ker \ell$. Then*

$$\beta(q) = 4\operatorname{Arf}(Q|_H) + \begin{cases} 1, & Q(w) = 0, \\ 7, & Q(w) = 1 \end{cases} \pmod 8.$$

Proof. In even dimension, translation by a gives $W(Q+\ell) = (-1)^{Q(a)} W(Q)$, while $W(Q) = (-1)^{\operatorname{Arf}(Q)} 2^{\dim V/2}$; substitute these in (9). In odd dimension, sum over the pairs $h, h+w$ for $h \in H$. The Gauss sum factors as

$$(-1)^{\operatorname{Arf}(Q|_H)} 2^{(\dim V - 1)/2} \big(1 + i(-1)^{Q(w)}\big),$$

whose two possible final factors have octants 1 and 7. □

For every ordinary quadratic form R on V, let $\mathcal{A}_R(x)$ be the finite normal-play arena of Theorem 8. The only interface used here is

$$o(\mathcal{A}_R(x)) \in \{\mathcal{P}, \mathcal{N}\}, \qquad o(\mathcal{A}_R(x)) = \mathcal{P} \iff R(x) = 0.$$

In particular, no impartiality assertion about this arena is needed.

Theorem 10 (intrinsic Brown selector). *For $q = \widehat{\ell} + 2Q$, the single partizan game*

$$\mathcal{B}_q(x) = \{\mathcal{A}_{Q+\ell}(x) \mid \mathcal{A}_Q(x)\}$$

has the outcome table

$q(x)$	$\ell(x)$	$Q(x)$	$Q(x)+\ell(x)$	$o(\mathcal{A}_{Q+\ell}(x))$	$o(\mathcal{A}_Q(x))$	$o(\mathcal{B}_q(x))$
0	0	0	0	$\mathcal{P}$	$\mathcal{P}$	$\mathcal{N}$
1	1	0	1	$\mathcal{N}$	$\mathcal{P}$	$\mathcal{R}$
2	0	1	1	$\mathcal{N}$	$\mathcal{N}$	$\mathcal{P}$
3	1	1	0	$\mathcal{P}$	$\mathcal{N}$	$\mathcal{L}$.

Thus the fixed decoder $\mathcal{N}, \mathcal{R}, \mathcal{P}, \mathcal{L} \mapsto 0, 1, 2, 3$ recovers $q(x)$.

Proof. If Left starts, her sole move enters $\mathcal{A}_{Q+\ell}(x)$ with Right to move, so Left wins exactly when $Q(x) + \ell(x) = 0$. If Right starts, his sole move enters $\mathcal{A}_Q(x)$ with Left to move, so Right wins exactly when $Q(x) = 0$. These two starter bits give the displayed four rows. □

The four outcome classes are, by definition, the two truth values ($G \geq 0, G \leq 0$): $\mathcal{P} = (1, 1)$, $\mathcal{L} = (1, 0)$, $\mathcal{R} = (0, 1)$, and $\mathcal{N} = (0, 0)$. Hence the selector packages the correlated pair $(Q + \ell, Q)$ into the two starter faces of one game tree; it is not an additional algebraic invariant. The decoder is only a label bijection, and outcomes do not compose under disjunctive sum. Indeed, if the followers are replaced by the simplest games with their stated outcomes, then $\mathcal{B}_q(x)$ is respectively

$$*, \quad \downarrow, \quad 0, \quad \uparrow \qquad (q(x) = 0, 1, 2, 3),$$

and $\uparrow + \downarrow = 0 \neq *$ displays the failure of additivity. The construction does not synchronize two plays: after the root move, exactly one follower is entered and ordinary normal play determines the winner.

Corollary 2 (outcome census). *If $n_{\mathcal{C}}$ counts the inputs with selector outcome $\mathcal{C}$, then*

$$G(q) = n_{\mathcal{N}} + in_{\mathcal{R}} - n_{\mathcal{P}} - in_{\mathcal{L}}.$$

Thus the four-outcome census retains exactly the correlated Walsh data and, when defined, the Brown phase.

Proof. Group the terms $i^{q(x)}$ by the table in Theorem 10. □

Remark 13 (the global boundary). This is a finite-space construction. Ambient-coherent Brown data on all short-game values vanish by two-divisibility; the stronger game-exterior statement and its exact coherence hypothesis are recorded in Appendix A. The section/framing issue for a bare $\mathbb{Z}/4$ central extension is separate; its algebraic core is kernel-checked in `formal/Ogdoad/BrownGame.lean`.

13 Conclusion

Every quadratic refinement of every finite alternating binary space therefore has a uniform bounded-access normal-play realization. A public Witt frame makes the polar graph a matching, weighted source pairs preserve original-frame singleton locality, the safe-front theorem forces the exact charge in every dimension, and the phase-aware claim compiles that charge to a P/N outcome. For the motivating Gold trace forms, nimber operations source the diagonal and the Arf invariant with radical data gives the resulting zero-set bias.

The surrounding negative results make the access boundary sharp. Linear coin-turning reaches the lexicode floor but not a genuine quadratic zero set; the extraspecial extension identifies the missing noncommutative datum; symmetric, commutative, and bounded-total-observation architectures cannot carry it uniformly; and extensional fork tests cannot define naturality. Weighted-source Witt–FIFO meets the necessary observation span with singleton support. The remaining arbitrary-graph FIFO contraction is a separate combinatorial conjecture, not a gap in the Gold theorem.

A The game-exterior obstruction

Altman and Lipparini ask whether Clifford algebras can be constructed game-theoretically [1, Problem 5.3(j)]. Their Remark 3.5 also isolates the basic boundary: Conway equivalence makes games an abelian group under disjunctive sum, but does not make Conway product well-defined on arbitrary game values. The theorem below gives a negative answer to that question for a precise and natural Clifford interpretation: an additive grade-one realization with coefficient-faithful square and polar relations that is coherent under adjoining short-game roots. It does not rule out a different multiplication or a different naturality contract.

Write ShUg for the additive group of short partizan games. An ambient-coherent datum must mean either one datum on all of ShUg, or a compatible system over *every* finitely generated subgroup (equivalently, over a cofinal directed subsystem). In the latter formulation, inclusion maps must inject the coefficient rings and algebras and preserve the grade-one maps, squares, and anticommutators. The filtered colimits of those rings, algebras, and groups glue such a system to the former global datum, and restriction gives the converse. This quantifier is essential: a family confined to selected subgroups need not see an actual root. In particular, the nonzero Gold forms on finite nimber subgroups would otherwise be counterexamples to the theorem.

Let R be a commutative ring, C an R-algebra, and $\iota\colon R \hookrightarrow C$ an injective coefficient map. A coefficient-faithful datum consists of an additive map $i\colon \mathrm{ShUg} \to C$ and functions Q, B satisfying

$$i(x)^2 = \iota(Q(x)), \qquad i(x)i(y) + i(y)i(x) = \iota(B(x,y)).$$

The equations and additivity already force the usual polarization and biadditivity identities. In particular, the diagonal anticommutator gives

$$\iota(B(y,y)) = 2i(y)^2 = \iota(2Q(y)), \qquad B(y,y) = 2Q(y), \tag{10}$$

and hence $Q(2y) = 4Q(y)$.

Theorem 11 (ambient-coherent game-exterior obstruction)**.** *In every ambient-coherent coefficient-faithful Clifford datum on* ShUg*,*

$$Q(x),\ B(x,y) \ \in \bigcap_{k\geq 0} 4^k R \qquad (x, y \in \mathrm{ShUg}).$$

Moreover, for every torsion game t and every short game x,

$$Q(t) = 0, \qquad B(t,x) = B(x,t) = 0.$$

Consequently the quadratic datum factors through the torsion-free quotient.

Proof. Moews's proof of his structure theorem uses Norton halving, and gives 2-divisibility of ShUg; all finite orders are powers of 2 by Conway's Theorem 92 [10, Theorem 92], as quoted by Moews [28]. Thus, for every k, write $x = 2^k u$ and $y = 2^k v$. Iterating (10) gives

$$Q(x) = 4^k Q(u), \qquad B(x, y) = 4^k B(u, v),$$

where the second equality uses the forced biadditivity of B. This proves the intersection assertion.

For the torsion assertion, let $nt = 0$, with n a power of 2. Choose y with $ny = t$. Since $ni(t) = i(nt) = 0$,

$$i(t)^2 = i(t)i(ny) = (ni(t))i(y) = 0,$$

so coefficient faithfulness gives $Q(t) = 0$. To kill the polar value, choose z with $nz = x$; no root of t is needed here:

$$\begin{aligned} i(t)i(x) + i(x)i(t) &= i(t)i(nz) + i(nz)i(t) \\ &= (ni(t))i(z) + i(z)(ni(t)) = 0. \end{aligned}$$

Injectivity gives $B(t, x) = 0$, and symmetry gives $B(x, t) = 0$. □

The diagonal identity in (10) is where coefficient faithfulness is indispensable: it equates two elements of C, then uses injectivity of ι to recover an equality in R. The torsion anticommutator argument itself uses only additivity of i and bilinearity of multiplication.

Corollary 3 (coefficient consequences)**.** *Over $\mathbb{Z}$, in characteristic 2, or over $\mathbb{Z}/4$, the quadratic and polar data are identically zero. In every coefficient ring, all torsion generators are exterior/nilpotent and polar-orthogonal to every grade-one generator.*

Proof. In each of the three named targets, $\bigcap_{k\geq 0} 4^k R = 0$. The remaining statement is the torsion part of Theorem 11. □

There is a concrete root behind the abstract input. In Moews's notation, if N_H is the least nonnegative integer with $H + 2N_H > 0$, Norton multiplication gives

$$\phi(H) = \tfrac{1}{2} \cdot (H + 2N_H) - N_H, \qquad 2\phi(H) = H.$$

For $H = *$, $N_* = 1$, so

$$\phi(*) = \tfrac{1}{2} \cdot (* + 2) - 1, \qquad 2\phi(*) = (* + 2) - 2 = *.$$

Indeed, the defining Norton rule is additive in its real multiplier, so $2(\frac{1}{2} \cdot U) = 1 \cdot U = U$; this verifies the displayed halving identity directly. This is the step used inside Moews's proof of Theorem 9, and is independently recorded in [24, Corollary 6.2.5].

What remains. Over $\mathbb{Z}$ even the torsion-free dyadic directions admit no nonzero additive scalar readout, since $\mathrm{Hom}(\mathbb{Z}[1/2], \mathbb{Z}) = 0$. The mean-value homomorphism does survive after passing to a target containing the relevant dyadic values (in particular, one in which 2 is invertible); its square and polarization give the usual free-direction examples. Nor is the obstruction a claim about every quotient: for example,

$$\mathbb{Z}[1/2]/3\mathbb{Z}[1/2] \cong \mathbb{Z}/3$$

carries ordinary quadratic data over $\mathbb{F}_3$. These elementary divisible/torsion distinctions belong to the standard abelian-group calculus [19, 25]; Wall's quadratic and linking forms on finite groups are the classical contrasting setting [36].

Three escapes are therefore real but deliberately outside the theorem. First, a hand-supplied table on a root-incomplete subgroup is the existing engineering API. Second, the Gold forms in the nimber core are game-native for a different, operation-based naturality condition: they use that core's addition, nim-product, Frobenius, and trace. Requiring extension across all short-game inclusions is one naturality condition, not the weakest possible one; it is chosen here because it asks whether a value-level Clifford datum survives adjoining the actual partizan roots of its generators. Under that condition those Gold forms cannot extend. Finally, a construction without an additive grade-one map is not constrained by this theorem, but is no longer the Clifford datum just defined.

The algebraic root-collapse and coefficient-faithfulness core is kernel-checked in `formal/Ogdoad/GameExterior.lean`. The passage from a cofinal family to its colimit, and the sharpened $\bigcap_k 4^k R$ conclusion, are paper-level arguments here.

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